

Project Title
Dictionary-Based Misspelled Word Detection System

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

in

A.Y. 2026-2027

Submitted by

Hansika Reddy Gunapati (2520030237)

Adoori Teja Hasini Reddy (2520030048)

Anthony Nissy (2520030064)

Team Number: 16

Section: 7

Under the guidance of

Dr. S. Madhavi

CSE



Hyderabad-500090, Telangana, India.

ABSTRACT

The Dictionary-Based Misspelled Word Detection System is a text-processing application designed to identify, flag, and correct misspelled words in documents, log files, and user-generated input using efficient string matching algorithms. As the volume of digital text grows across platforms, ensuring accuracy and readability has become essential for effective communication and reliable downstream processing such as search, analytics, and natural language understanding.

Traditional spellchecking approaches often rely on naïve, character-by-character scanning or limited, static word lists, making them slow and inefficient when applied to large-scale text datasets. This creates a need for a system that can validate text against a comprehensive dictionary quickly and accurately, while also scaling to handle real-world data that includes diverse vocabulary, mixed languages, and Unicode characters. The proposed system addresses this by comparing input text against a predefined dictionary using efficient string matching structures such as Tries, the Aho–Corasick algorithm, and suffix arrays, enabling fast lookup and pattern detection even across large volumes of text.

Once a misspelled word is located, the system suggests likely corrections using edit distance and phonetic matching techniques, and compiles a report summarizing the errors found along with their suggested fixes. Key features include dictionary-based lookup, multilingual and Unicode support, automated correction suggestions, and performance benchmarking against naïve spellchecking methods to demonstrate the efficiency gained through optimized string matching.

By combining accurate detection with fast processing, the system minimizes misinterpretation caused by spelling errors, improves text readability, and supports practical applications in text analytics, natural language processing, and automated document review, making it valuable for both individual users and large-scale data pipelines. The system is designed to be lightweight and easily integrable into existing text-processing pipelines, requiring minimal setup beyond loading the dictionary. Future enhancements may include context-aware correction using n-gram language models to better handle words that are spelled correctly but used incorrectly. Overall, the project demonstrates how classical data structures and algorithms remain highly relevant to solving everyday real-world text quality problems.

Signature of the Guide